

SHM Worked Example 5.19

Reference: Applied Mechanics: Reed's Vol. 2

Ex. 5.19: A helical spring with a mass of 6 kg has a mass of 10 kg attached to its free end. The mass is pulled down 40 mm from the equilibrium position and then released.

Given:

- Mass of the spring, $m_s = 6$ kg
- Mass attached, $m = 10$ kg
- Spring stiffness, $k = 532$ N/m
- Initial displacement (amplitude), $A = 40$ mm = 0.0400 m
- Position to find kinetic energy, $x = 30$ mm = 0.0300 m

(i) Frequency of Oscillation

The natural frequency of a mass-spring system is given by:

$$\omega_n = \sqrt{\frac{k}{m_{\text{eff}}}}$$

where the effective mass accounts for the distributed mass of the spring:

$$m_{\text{eff}} = m + \frac{m_s}{3}$$

Substituting the given values:

$$m_{\text{eff}} = 10 + \frac{6}{3} = 12 \text{ kg}$$

Calculating the angular frequency:

$$\omega_n = \sqrt{\frac{532}{12}} = \sqrt{44.3333} = 6.6583 \text{ rad/s}$$

The frequency of oscillation is:

$$f = \frac{\omega_n}{2\pi} = \frac{6.6583}{2\pi} = 1.0597 \text{ Hz}$$

(ii) Kinetic Energy at $x = 30$ mm

The velocity of the attached mass at displacement x is:

$$v = \omega_n \sqrt{A^2 - x^2} = 6.6583 \times \sqrt{(0.0400)^2 - (0.0300)^2}$$

$$v = 6.6583 \times \sqrt{0.0007} = 6.6583 \times 0.0265 = 0.1762 \text{ m/s}$$

The kinetic energy of the attached mass is:

$$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 10 \times (0.1762)^2 = 0.1552 \text{ J}$$

Final Answers

- **Frequency of oscillation:** $f = 1.0597$ Hz
- **Kinetic energy at $x = 30$ mm:** $KE = 0.1552$ J